



MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of faculty: Kalyan Singh

Subject & Subject Code: ELEMENT OF MECHANICAL ENGINEERING,

Semester: I

Session: 2021-22 (Odd Sem.)

Branch: CE

Batch: E

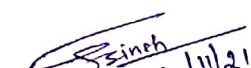
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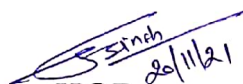
Date of Submission: 20/11/2021

Lecture No.	Topics Covered	Total Page
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References:

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons


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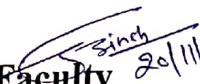
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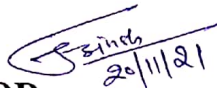
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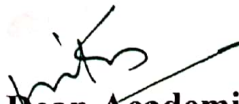
Lecture No.	Topics Covered	Plagiarism %
33	Introduction and types of Belt	35
34	Length of Open Belt	46
35	Length of Cross Belt Drive	51
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37	Introduction and Types of Ropes	49
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40	Velocity Ratio for various types of Gear Trains	45
	Average plagiarism %	48.25%

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LECTURE -33

Belt and Rope Drives

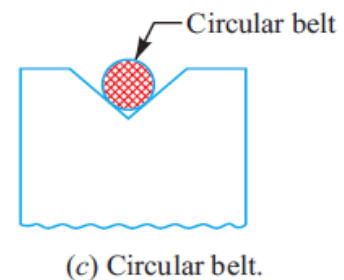
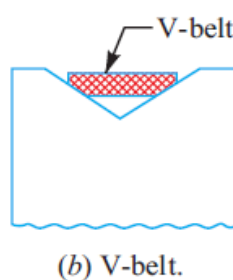
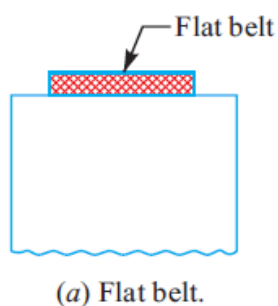
1. Introduction

The belts or ropes are used to transmit power from one shaft to another by means of pulleys which rotate at the same speed or at different speeds.



2. Types of Belts

1. **Flat belt** is mostly used in the factories and workshops, where a moderate amount of power is to be transmitted, from one pulley to another when the two pulleys are not more than 8 metres apart.
2. **V-belt** is mostly used in the factories and workshops, where a moderate amount of power is to be transmitted, from one pulley to another, when the two pulleys are very near to each other.
3. **Circular belt or rope** is mostly used in the factories and workshops, where a great amount of power is to be transmitted, from one pulley to another, when the two pulleys are more than 8 meters apart.



2. Types of Flat Belt Drives

The power from one pulley to another may be transmitted by any of the following types of belt drives:

1. **Open belt drive.** The open belt drive, as shown in Fig. 1, is used with shafts arranged parallel and rotating in the same direction. The tension in tight side is more than the tension in slack side.
2. **Crossed or twist belt drive.** The crossed or twist belt drive, as shown in Fig.2, is used with shafts arranged parallel and rotating in the opposite directions.

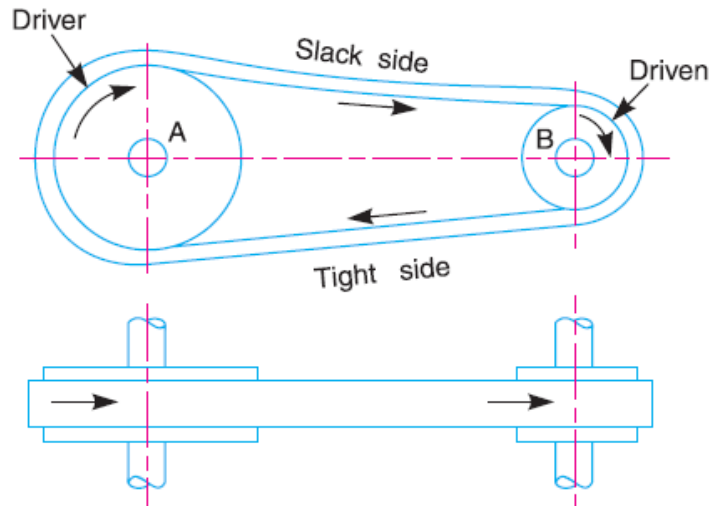


Fig. 1. Open belt drive.

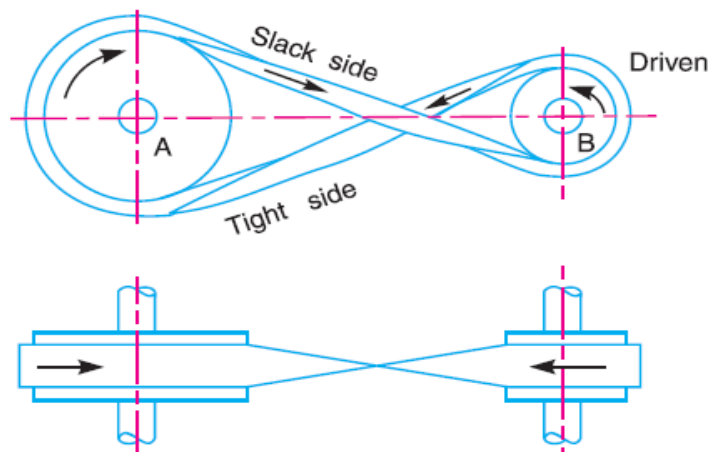


Fig. 2. Crossed or twist belt drive.

In order to avoid the wear in the belt intersection the shafts should be placed at a maximum distance of $20b$, where b is the width of belt and the speed of the belt should be less than 15 m/s.

3. **Quarter turn belt drive.** The quarter turn belt drive also known as right angle belt drive, as shown in Fig. 3, is used with shafts arranged at right angles and rotating in one definite direction.

4. **Belt drive with idler pulleys.** A belt drive with an idler pulley, as shown in Fig.4, is used with shafts arranged parallel and when an open belt drive cannot be used due to small angle of contact on the smaller pulley. This type of drive is provided to obtain high velocity ratio and when the required belt tension cannot be obtained by other means.

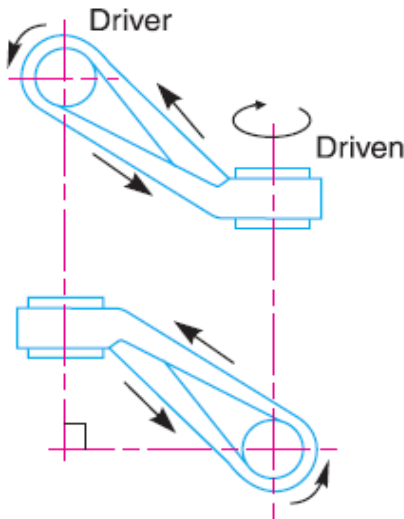


Fig. 3 Quarter turn belt drive.

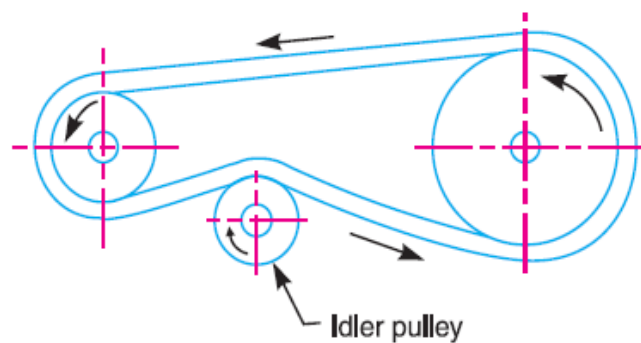


Fig. 4 Belt drive with single idler pulley.

5. **Compound belt drive.** A compound belt drive, as shown in Fig. 5, is used when power is transmitted from one shaft to another through a number of pulleys.

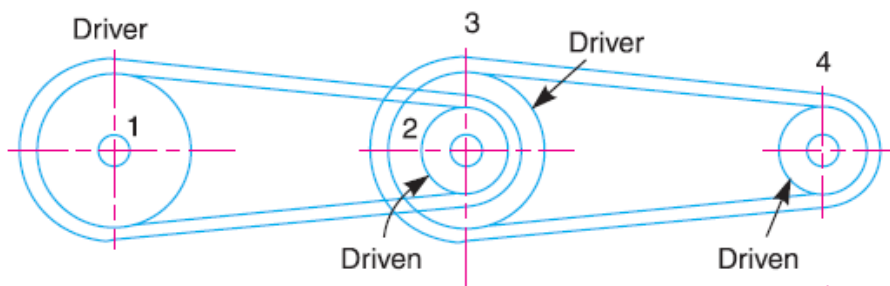


Fig .5. Compound belt drive.

6. **Stepped or cone pulley drive.** A stepped or cone pulley drive, as shown in Fig.6, is used for changing the speed of the driven shaft while the main or driving shaft runs at constant speed. This is accomplished by shifting the belt from one part of the steps to the other.
7. **Fast and loose pulley drive.** A fast and loose pulley drive, as shown in Fig. 7, is used when the driven or machine shaft is to be started or stopped when ever

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desired without interfering with the driving shaft. A pulley which is keyed to the machine shaft is called **fast pulley** and runs at the same speed as that of machine shaft. A loose pulley runs freely over the machine shaft and is incapable of transmitting any power. When the driven shaft is required to be stopped, the belt is pushed on to the loose pulley by means of sliding bar having belt forks.

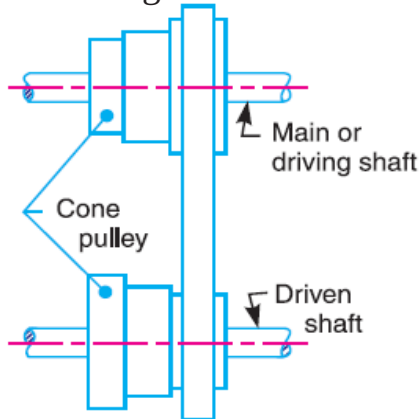


Fig. 6. Stepped or cone pulley drive.

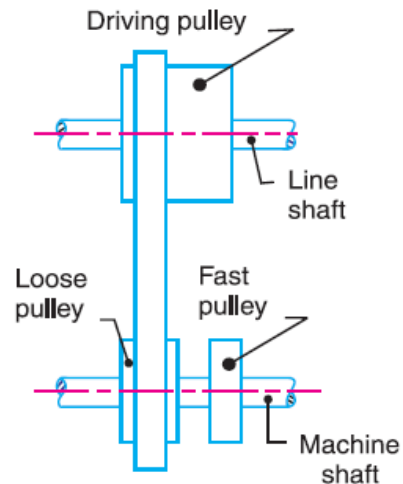


Fig. 7. Fast and loose pulley drive.

3. Velocity Ratio of Belt Drive

It is the **ratio between the velocities of the driver and the follower or driven**. It may be expressed, mathematically, as:

$$\frac{N_2}{N_1} = \frac{d_1}{d_2}$$

Where d_1 = Diameter of the driver,

d_2 = Diameter of the follower,

N_1 = Speed of the driver in r.p.m., and

N_2 = Speed of the follower in r.p.m.

When the thickness of the belt (t) is considered, then velocity ratio,

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t}$$

4. Velocity Ratio of a Compound Belt Drive

Sometimes the power is transmitted from one shaft to another, through a number of pulleys as shown in Fig. 5.

We know that velocity ratio of pulleys 1 and 2,

$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \quad i$$

Similarly, velocity ratio of pulleys 3 and 4,

$$\frac{N_4}{N_3} = \frac{d_3}{d_4} \quad ii$$

Multiplying equations (i) and (ii),

$$\frac{N_4}{N_1} = \frac{d_1 \times d_3}{d_2 \times d_4} \quad (N_2 = N_3, \text{ because of the same shaft})$$

or
$$\frac{\text{Speed of last driven}}{\text{Speed of first driver}} = \frac{\text{Product of diameters of drivers}}{\text{Product of diameters of driven}}$$

5. Slip of Belt

In the previous articles, we have discussed the motion of belts and shafts assuming a firm frictional grip between the belts and the shafts. But sometimes, the frictional grip becomes insufficient. This may cause some forward motion of the driver without carrying the belt with it. This may also cause some forward motion of the belt without carrying the driven pulley with it. This is called *slip of the belt* and is generally expressed as a percentage.

The result of the belt slipping is to reduce the velocity ratio of the system. As the slipping of the belt is a common phenomenon, thus the belt should never be used where a definite velocity ratio is of importance (as in the case of hour, minute and second arms in a watch).

Let $s_1 \%$ = Slip between the driver and the belt, and

$s_2 \%$ = Slip between the belt and the follower.

Velocity of the belt passing over the driver per second

$$v = \frac{\pi d_1 N_1}{60} - \frac{\pi d_1 N_1}{60} \times \frac{s_1}{100} = \frac{\pi d_1 N_1}{60} \left(1 - \frac{s_1}{100}\right)$$

and velocity of the belt passing over the follower per second,

$$\frac{\pi d_2 N_2}{60} = v - v \times \frac{s_2}{100} = v \left(1 - \frac{s_2}{100}\right)$$

$$\begin{aligned}\frac{\pi d_2 N_2}{60} &= \frac{\pi d_1 N_1}{60} \left(1 - \frac{s_1}{100}\right) \left(1 - \frac{s_2}{100}\right) \\ \frac{N_2}{N_1} &= \frac{d_1}{d_2} \left(1 - \frac{s_1}{100} - \frac{s_2}{100}\right) \quad \left(\text{neglecting } \frac{s_1 \times s_2}{100 \times 100}\right) \\ &= \frac{d_1}{d_2} \left(1 - \frac{s_1 + s_2}{100}\right) = \frac{d_1}{d_2} \left(1 - \frac{s}{100}\right)\end{aligned}$$

Where $s = s_1 + s_2$ total percentage of slip.

If thickness of the belt (t) is considered, then

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \left(1 - \frac{s}{100}\right)$$

6. Creep of Belt

When the belt passes from the slack side to the tight side, a certain portion of the belt extends and it contracts again when the belt passes from the tight side to slack side. Due to these changes of length, there is a relative motion between the belt and the pulley surfaces. This relative motion is termed as **creep**. The total effect of creep is to reduce slightly the speed of the driven pulley or follower. Considering creep, the velocity ratio is given by

$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \times \frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}$$

Where σ_1 and σ_2 = Stress in the belt on the tight and slack side respectively, and
 E = Young's modulus for the material of the belt.

7. Length of an Open Belt Drive

In this case, both the pulleys rotate in the **same** direction.

Let r_1 and r_2 = Radii of the larger and smaller pulleys,

X = Distance between the centers of two pulleys (i.e. $O_1 O_2$), and

L = Total length of the belt.

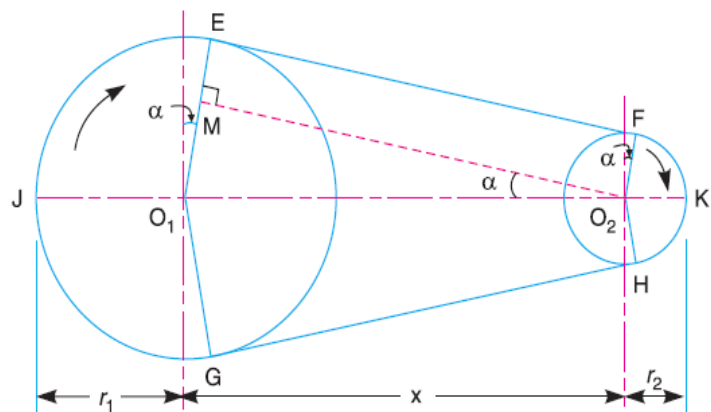


Fig. 8. Length of an open belt drive.

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Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H as shown in Fig. 10. Through O_2 , draw $O_2 M$ parallel to FE .

From the geometry of the figure, we find that $O_2 M$ will be perpendicular to $O_1 E$.

Let the angle $MO_2 O_1 = \alpha$ radians.

$$\begin{aligned} L &= \text{Arc } GJE + EF + \text{Arc } FKH + HG \\ &= 2 (\text{Arc } JE + EF + \text{Arc } FK) \end{aligned}$$

From the geometry of the figure, we find that

$$\sin \alpha = \frac{O_1 M}{O_1 O_2} = \frac{O_1 E - EM}{O_1 O_2} = \frac{r_1 - r_2}{x}$$

Since α is very small, therefore putting

$$\sin \alpha = \alpha \text{ (in radians)} = \frac{r_1 - r_2}{x}$$

$$\therefore \text{Arc } JE = r_1 \left(\frac{\pi}{2} + \alpha \right)$$

$$\text{Similarly Arc } FK = r_2 \left(\frac{\pi}{2} - \alpha \right)$$

and

$$\begin{aligned} EF &= MO_2 = \sqrt{(O_1 O_2)^2 - (O_1 M)^2} = \sqrt{x^2 - (r_1 - r_2)^2} \\ &= x \sqrt{1 - \left(\frac{r_1 - r_2}{x} \right)^2} \end{aligned}$$

Expanding this equation by binomial theorem,

$$EF = x \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x} \right)^2 + \dots \right] = x - \frac{(r_1 - r_2)^2}{2x}$$

$$L = 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + x - \frac{(r_1 - r_2)^2}{2x} + r_2 \left(\frac{\pi}{2} - \alpha \right) \right]$$

Since $\alpha = \frac{r_1 - r_2}{x}$ then

$$L = \pi (r_1 + r_2) + 2x + \frac{(r_1 - r_2)^2}{x}$$

1. Length of a Cross Belt Drive

In this case, both the pulleys rotate in **opposite** directions as shown in Fig. 1.

Let r_1 and r_2 = Radii of the larger and smaller pulleys,

X = Distance between the centers of two pulleys (i.e. $O_1 O_2$), and

L = Total length of the belt.

Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H . Through O_2 , draw O_2M parallel to FE . From the geometry of the figure, we find that O_2M will be perpendicular to O_1E .

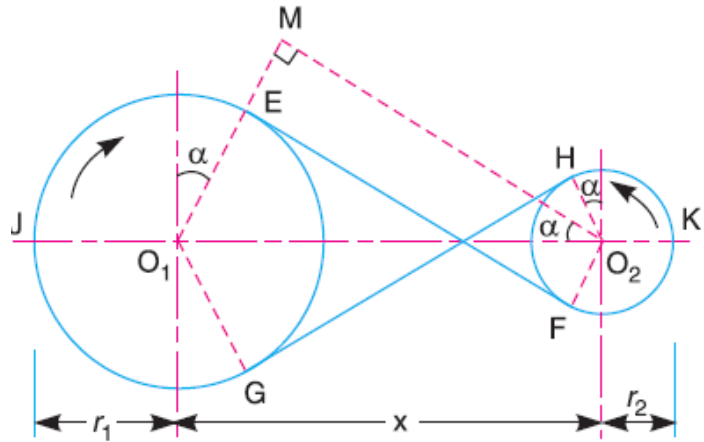


Fig. 1. Length of a cross belt drive.

Let the angle $MO_2O_1 = \alpha$ radians.

$$\begin{aligned} L &= \text{Arc } GJE + EF + \text{Arc } FKH + HG \\ &= 2 (\text{Arc } JE + EF + \text{Arc } FK) \end{aligned}$$

From the geometry of the figure, we find that

$$\sin \alpha = \frac{O_1 M}{O_1 O_2} = \frac{O_1 E + EM}{O_1 O_2} = \frac{r_1 + r_2}{x}$$

Since α is very small, therefore putting

$$\sin \alpha = \alpha \text{ (in radians)} = \frac{r_1 + r_2}{x}$$

$$\therefore \text{Arc } JE = r_1 \left(\frac{\pi}{2} + \alpha \right)$$

$$\text{Similarly } \text{Arc } FK = r_2 \left(\frac{\pi}{2} + \alpha \right)$$

$$EF = MO_2 = \sqrt{(O_1 O_2)^2 - (O_1 M)^2} = \sqrt{x^2 - (r_1 + r_2)^2}$$

$$= x \sqrt{1 - \left(\frac{r_1 + r_2}{x} \right)^2}$$

Expanding this equation by binomial theorem,

$$EF = x \left[1 - \frac{1}{2} \left(\frac{r_1 + r_2}{x} \right)^2 + \dots \right] = x - \frac{(r_1 + r_2)^2}{2x}$$

$$L = 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + x - \frac{(r_1 + r_2)^2}{2x} + r_2 \left(\frac{\pi}{2} + \alpha \right) \right]$$

Substituting the value of $\alpha = \frac{r_1 + r_2}{x}$ then

$$L = \pi (r_1 + r_2) + 2x + \frac{(r_1 + r_2)^2}{x}$$

2. Power Transmitted by a Belt

Let T_1 and T_2 = Tensions in the tight and slack side of the belt respectively in N,
 r_1 and r_2 = Radii of the driver and follower respectively, and
 v = Velocity of the belt in m/s.

$\therefore$ Work done per second = $(T_1 - T_2) v$ N-m/s

and power transmitted, $P = (T_1 - T_2) v$ Watt

3. Ratio of Driving Tensions For Flat Belt Drive

Consider a driven pulley rotating in the clockwise direction as shown in Fig. 2.

Let T_1 = Tension in the belt on the tight side,

T_2 = Tension in the belt on the slack side, and

θ = Angle of contact in radians.

Now consider a small portion of the belt PQ , subtending an angle $\delta\theta$ at the centre of the pulley. The belt PQ is in equilibrium under the following forces:

1. Tension T in the belt at P ,
2. Tension $(T + \delta T)$ in the belt at Q ,
3. Normal reaction R_N , and
4. Frictional force, $F = \mu \times R_N$,

where μ is the coefficient of friction between the belt and pulley.

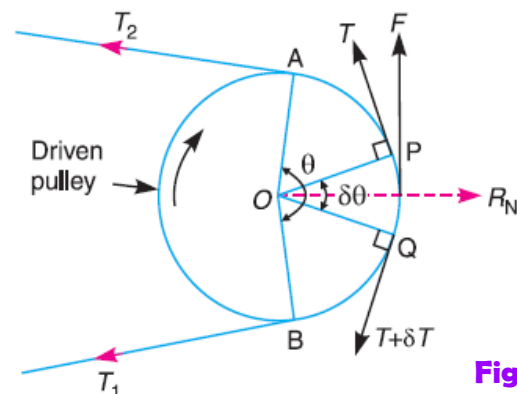


Fig .2

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$$\Sigma F_x = 0$$

$$R_N = (T + \delta T) \sin \frac{\delta \theta}{2} + T \sin \frac{\delta \theta}{2}$$

Since the angle $\delta \theta$ is very small, therefore putting $\sin \delta \theta / 2 = \delta \theta / 2$ in equation

$$R_N = (T + \delta T) \frac{\delta \theta}{2} + T \times \frac{\delta \theta}{2} = \frac{T \cdot \delta \theta}{2} + \frac{\delta T \cdot \delta \theta}{2} + \frac{T \cdot \delta \theta}{2} = T \cdot \delta \theta$$

... $\left(\text{Neglecting } \frac{\delta T \cdot \delta \theta}{2} \right)$

$$\Sigma F_y = 0$$

$$\mu \times R_N = (T + \delta T) \cos \frac{\delta \theta}{2} - T \cos \frac{\delta \theta}{2}$$

Since the angle $\delta \theta$ is very small, therefore putting $\cos \delta \theta / 2 = 1$ in equation

$$\mu \times R_N = T + \delta T - T = \delta T \quad \text{or} \quad R_N = \frac{\delta T}{\mu}$$

$$T \cdot \delta \theta = \frac{\delta T}{\mu} \quad \text{or} \quad \frac{\delta T}{T} = \mu \cdot \delta \theta$$

$$\int_{T_2}^{T_1} \frac{\delta T}{T} = \mu \int_0^\theta \delta \theta \quad \text{or} \quad \ln \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta \quad \text{or} \quad \frac{T_1}{T_2} = e^{\mu \cdot \theta}$$

4. Determination of Angle of Contact

The angle of contact or lap in open belt drives:

$$\theta = (180^\circ - 2\alpha) \frac{\pi}{180} \text{ rad} \quad \sin \alpha = \frac{r_1 - r_2}{x}$$

The angle of contact or lap in crossed belt drives:

$$\theta = (180^\circ + 2\alpha) \frac{\pi}{180} \text{ rad} \quad \sin \alpha = \frac{r_1 + r_2}{x}$$

5. Maximum Tension in the Belt

The maximum tension in the belt ($T_{\max}$) is equal to the total tension in the tight side of the belt (T_{t1}).

Let $\sigma_{\max}$ = Maximum safe stress in N/mm^2 ,

b = Width of the belt in mm, and

t = Thickness of the belt in mm.

We know that maximum tension in the belt,

$$T_{\max} = \sigma_{\max} \cdot b \cdot t$$

When centrifugal tension is neglected, then

$$T_{\max} = T_1$$

and when centrifugal tension is considered, then

$$T_{\max} = T_1 + T_C$$

6. Condition For the Transmission of Maximum Power

$$P = (T_1 - T_2) v \quad \dots (i)$$

$$\frac{T_1}{T_2} = e^{\mu \cdot \theta} \quad \text{or} \quad T_2 = \frac{T_1}{e^{\mu \cdot \theta}} \quad \dots (ii)$$

Substituting the value of T_2 in equation (i),

$$P = \left(T_1 - \frac{T_1}{e^{\mu \cdot \theta}} \right) v = T_1 \left(1 - \frac{1}{e^{\mu \cdot \theta}} \right) v = T_1 \cdot v \cdot C \quad \dots (iii)$$

Where $C = 1 - \frac{1}{e^{\mu \cdot \theta}}$

But $T_1 = T_{\max} - T_C$

$$P = (T_{\max} - T_C) v \cdot C$$

$$= (T_{\max} - m \cdot v^2) v \cdot C = (T_{\max} v - m v^3) C \quad \dots \text{(Substituting } T_C = m \cdot v^2 \text{)}$$

For maximum power, differentiate the above expression with respect to v and

equate to zero, $\frac{dP}{dv} = 0 \quad \therefore T_{\max} - 3 m \cdot v^2 = 0$

$$T_{\max} - 3 T_C = 0 \quad \text{or} \quad T_{\max} = 3 T_C$$

Note: We know that $T_1 = T_{\max} - T_C$ and for maximum power, $T_C = \frac{T_{\max}}{3}$

$$T_1 = T_{\max} - \frac{T_{\max}}{3} = \frac{2T_{\max}}{3}$$

Example: 1 A leather belt is required to transmit 7.5 kW from a pulley 1.2 m in diameter, running at 250 r.p.m. The angle embraced is 165° and the coefficient of friction between the belt and the pulley is 0.3. If the safe working stress for the leather belt is 1.5 MPa, density of leather 1 Mg/m^3 and thickness of belt 10 mm, determine the width of the belt taking centrifugal tension into account.

Solution:

$$v = \pi d \cdot N / 60 = \pi \times 1.2 \times 250 / 60 = 15.71 \text{ m/s}$$

power transmitted (P),

$$7500 = (T_1 - T_2) v = (T_1 - T_2) 15.71$$

$\therefore$

$$T_1 - T_2 = 7500 / 15.71 = 477.4 \text{ N}$$

We know that

$$\frac{T_1}{T_2} = e^{\mu \cdot \theta} = \frac{0.3 \times 2.88}{e} = 2.375$$

$$T_1 = 824.6 \text{ N, and } T_2 = 347.2 \text{ N}$$

We know that mass of the belt per metre length,

$$\begin{aligned} m &= \text{Area} \times \text{length} \times \text{density} = b \cdot t \cdot l \cdot \rho \\ &= b \times 0.01 \times 1 \times 1000 = 10 b \text{ kg} \end{aligned}$$

$\therefore$ Centrifugal tension,

$$T_C = m \cdot v^2 = 10 b (15.71)^2 = 2468 b \text{ N}$$

and maximum tension in the belt,

$$T_{\max} = \sigma \cdot b \cdot t = 1.5 \times 10^6 \times b \times 0.01 = 15\,000 b \text{ N}$$

We know that

$$T_{\max} = T_1 + T_C \text{ or } 15000 b = 824.6 + 2468 b$$

$$15\,000 b - 2468 b = 824.6 \text{ or } 12\,532 b = 824.6$$

$\therefore$

$$b = 824.6 / 12532 = 0.0658 \text{ m} = 65.8 \text{ mm} \text{ Ans.}$$

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3. Ratio of Driving Tensions For Flat Belt Drive

Consider a driven pulley rotating in the clockwise direction as shown in Fig. 2.

Let T_1 = Tension in the belt on the tight side,

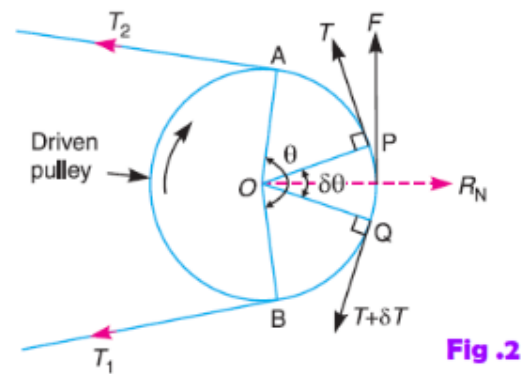
T_2 = Tension in the belt on the slack side, and

θ = Angle of contact in radians.

Now consider a small portion of the belt PQ , subtending an angle $\delta\theta$ at the centre of the pulley. The belt PQ is in equilibrium under the following forces:

1. Tension T in the belt at P ,
2. Tension $(T + \delta T)$ in the belt at Q ,
3. Normal reaction R_N , and
4. Frictional force, $F = \mu \times R_N$,

where μ is the coefficient of friction between the belt and pulley.



$$\Sigma F_x = 0$$

$$R_N = (T + \delta T) \sin \frac{\delta\theta}{2} + T \sin \frac{\delta\theta}{2}$$

Since the angle $\delta\theta$ is very small, therefore putting $\sin \delta\theta / 2 = \delta\theta / 2$ in equation

$$R_N = (T + \delta T) \frac{\delta\theta}{2} + T \times \frac{\delta\theta}{2} = \frac{T \cdot \delta\theta}{2} + \frac{\delta T \cdot \delta\theta}{2} + \frac{T \cdot \delta\theta}{2} = T \cdot \delta\theta$$

... (Neglecting $\frac{\delta T \cdot \delta\theta}{2}$)

$$\Sigma F_y = 0$$

$$\mu \times R_N = (T + \delta T) \cos \frac{\delta \theta}{2} - T \cos \frac{\delta \theta}{2}$$

Since the angle $\delta \theta$ is very small, therefore putting $\cos \delta \theta / 2 = 1$ in equation

$$\mu \times R_N = T + \delta T - T = \delta T \quad \text{or} \quad R_N = \frac{\delta T}{\mu}$$

$$T \cdot \delta \theta = \frac{\delta T}{\mu} \quad \text{or} \quad \frac{\delta T}{T} = \mu \cdot \delta \theta$$

$$\int_{T_2}^{T_1} \frac{\delta T}{T} = \mu \int_0^\theta \delta \theta \quad \text{or} \quad \ln \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta \quad \text{or} \quad \frac{T_1}{T_2} = e^{\mu \cdot \theta}$$

4. Determination of Angle of Contact

The angle of contact or lap in open belt drives:

$$\theta = (180^\circ - 2\alpha) \frac{\pi}{180} \text{ rad} \quad \sin \alpha = \frac{r_1 - r_2}{x}$$

The angle of contact or lap in crossed belt drives:

$$\theta = (180^\circ + 2\alpha) \frac{\pi}{180} \text{ rad} \quad \sin \alpha = \frac{r_1 + r_2}{x}$$

5. Maximum Tension in the Belt

The maximum tension in the belt ($T_{\max}$) is equal to the total tension in the tight side of the belt (T_1).

Let $\sigma_{\max}$ = Maximum safe stress in N/mm^2 ,
 b = Width of the belt in mm, and
 t = Thickness of the belt in mm.

We know that maximum tension in the belt,

$$T_{\max} = \sigma_{\max} \cdot b \cdot t$$

When centrifugal tension is neglected, then

$$T_{\max} = T_1$$

and when centrifugal tension is considered, then

$$T_{\max} = T_1 + T_C$$

6. Condition For the Transmission of Maximum Power

$$P = (T_1 - T_2) v \quad \dots (i)$$

$$\frac{T_1}{T_2} = e^{\mu \cdot \theta} \quad \text{or} \quad T_2 = \frac{T_1}{e^{\mu \cdot \theta}} \quad \dots (ii)$$

Substituting the value of T_2 in equation (i),

$$P = \left(T_1 - \frac{T_1}{e^{\mu \cdot \theta}} \right) v = T_1 \left(1 - \frac{1}{e^{\mu \cdot \theta}} \right) v = T_1 \cdot v \cdot C \quad \dots (iii)$$

Where $C = 1 - \frac{T_1}{e^{\mu \cdot \theta}}$

But $T_1 = T_{\max} - T_C$

$$P = (T_{\max} - T_C) v \cdot C$$

$$= (T_{\max} - m \cdot v^2) v \cdot C = (T_{\max} v - m v^3) C \quad \dots \text{(Substituting } T_C = m \cdot v^2)$$

For maximum power, differentiate the above expression with respect to v and

equate to zero, $\frac{dP}{dv} = 0 \quad \therefore T_{\max} - 3 m \cdot v^2 = 0$

$$T_{\max} - 3 T_C = 0 \quad \text{or} \quad T_{\max} = 3 T_C$$

Note: We know that $T_1 = T_{\max} - T_C$ and for maximum power, $T_C = \frac{T_{\max}}{3}$

$$T_1 = T_{\max} - \frac{T_{\max}}{3} = \frac{2T_{\max}}{3}$$

Example: 1 A leather belt is required to transmit 7.5 kW from a pulley 1.2 m in diameter, running at 250 r.p.m. The angle embraced is 165° and the coefficient of friction between the belt and the pulley is 0.3. If the safe working stress for the leather belt is 1.5 MPa, density of leather 1 Mg/m^3 and thickness of belt 10 mm, determine the width of the belt taking centrifugal tension into account.

Solution:

$$v = \pi d \cdot N / 60 = \pi \times 1.2 \times 250 / 60 = 15.71 \text{ m/s}$$

power transmitted (P),

$$7500 = (T_1 - T_2) v = (T_1 - T_2) 15.71$$

$$\therefore T_1 - T_2 = 7500 / 15.71 = 477.4 \text{ N}$$

We know that

$$\frac{T_1}{T_2} = e^{\mu \cdot \theta} = e^{0.3 \times 2.88} = 2.375$$

$$T_1 = 824.6 \text{ N, and } T_2 = 347.2 \text{ N}$$

We know that mass of the belt per metre length,

$$\begin{aligned} m &= \text{Area} \times \text{length} \times \text{density} = b \cdot t \cdot l \cdot \rho \\ &= b \times 0.01 \times 1 \times 1000 = 10 b \text{ kg} \end{aligned}$$

$\therefore$ Centrifugal tension,

$$T_C = m \cdot v^2 = 10 b (15.71)^2 = 2468 b \text{ N}$$

and maximum tension in the belt,

$$T_{\max} = \sigma \cdot b \cdot t = 1.5 \times 10^6 \times b \times 0.01 = 15000 b \text{ N}$$

$$\text{We know that } T_{\max} = T_1 + T_C \text{ or } 15000 b = 824.6 + 2468 b$$

$$15000 b - 2468 b = 824.6 \text{ or } 12532 b = 824.6$$

$$\therefore b = 824.6 / 12532 = 0.0658 \text{ m} = 65.8 \text{ mm} \text{ Ans.}$$



LECTURE -38**Syllabus :- GEARS AND GEAR TRAINS (EXCEPT EPICYCLIC GEAR TRAIN)****Introduction**

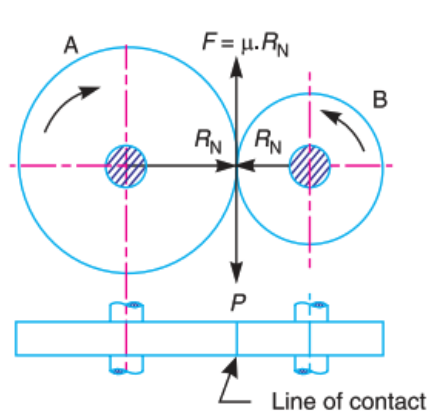
We have discussed about the slipping of a belt or rope which is a common phenomenon, in the transmission of motion or power between two shafts. The effect of slipping is to reduce the velocity ratio of the system. In precision machines, in which a definite velocity ratio is of importance (as in watch mechanism), the only positive drive is by means of gears or toothed wheels. A gear drive is also provided, when the distance between the driver and the follower is very small.

Friction Wheels

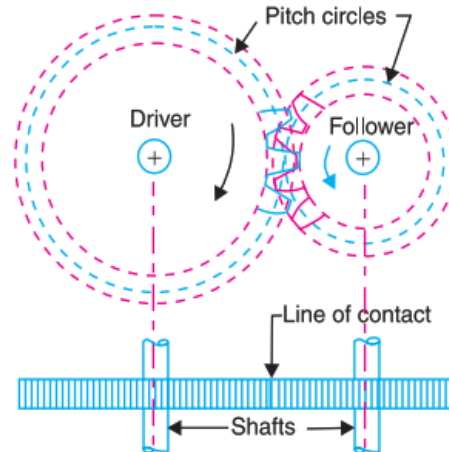
The motion and power transmitted by gears is kinematically equivalent to that transmitted by friction wheels or discs. In order to understand how the motion can be transmitted by two toothed wheels, consider two plain circular wheels A and B mounted on shafts, having sufficient rough surfaces and pressing against each other as shown in Fig. (a).

Let the wheel A be keyed to the rotating shaft and the wheel B to the shaft, to be rotated. A little consideration will show, that when the wheel A is rotated by a rotating shaft, it will rotate the wheel B in the opposite direction as shown in Fig. (a).

The wheel B will be rotated (by the wheel A) so long as the tangential force exerted by the wheel A does not exceed the maximum frictional resistance between the two wheels. But when the tangential force (P) exceeds the *frictional resistance (F), slipping will take place between the two wheels. Thus, the friction drive is not a positive drive.



(a) Friction wheels.



(b) Toothed wheels.

In order to avoid the slipping, a number of projections (called teeth) as shown in Fig. (b), are provided on the periphery of the wheel A, which will fit into the corresponding recesses on the periphery of the wheel B. A friction wheel with the teeth cut on it is known as toothed wheel or gear. The usual connection to show the toothed wheels is by their pitch circles.

Note: Kinematically, the friction wheels running without slip and toothed gearing are identical. But due to the possibility of slipping of wheels, the friction wheels can only be used for transmission of small powers.

Advantages and Disadvantages of Gear Drive

The following are the advantages and disadvantages of the gear drive as compared to belt, rope and chain drives:

Advantages

1. It transmits exact velocity ratio.
2. It may be used to transmit large power.
3. It has high efficiency.
4. It has reliable service.
5. It has compact layout.

Disadvantages

1. The manufacture of gears require special tools and equipment.
2. The error in cutting teeth may cause vibrations and noise during operation.

Classification of Toothed Wheels

The gears or toothed wheels may be classified as follows:

1. According to the position of axes of the shafts: The axes of the two shafts between which the motion is to be transmitted, may be

(a) Parallel, (b) Intersecting, and (c) Non-intersecting and non-parallel.

The two parallel and co-planar shafts connected by the gears is shown in above Fig. These gears are called spur gears and the arrangement is known as spur gearing. These gears have teeth parallel to the axis of the wheel as shown in above Fig. Another name given to the spur gearing is helical gearing, in which the teeth are inclined to the axis. The single and double helical gears connecting parallel shafts are shown in following Fig. (a) and (b) respectively. The double helical gears are known as herringbone gears. A pair of spur gears are kinematically equivalent to a pair of cylindrical discs, keyed to parallel shafts and having a line contact.

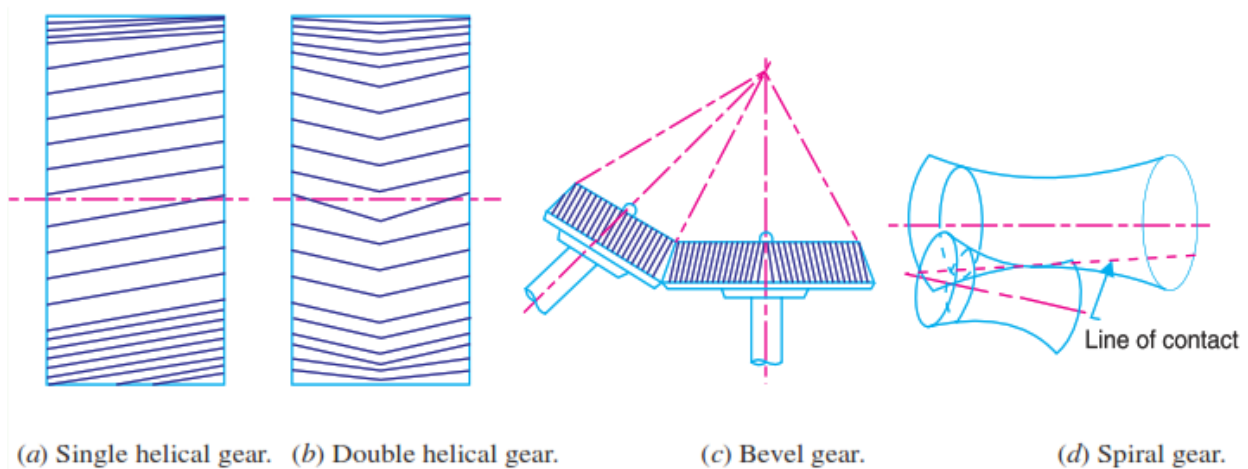
The two non-parallel or intersecting, but coplanar shafts connected by gears is shown in following Fig.(c). These gears are called bevel gears and the arrangement is known as bevel gearing. The bevel gears, like spur gears, may also have their teeth inclined to the face of the bevel, in which case they are known as helical bevel gears.

The two non-intersecting and non-parallel i.e. non-coplanar shaft connected by gears is shown in following Fig. (d). These gears are called skew bevel gears or spiral gears and the arrangement is known as skew bevel gearing or spiral gearing. This type of gearing also have a line contact, the rotation of which about the axes generates the two pitch surfaces known as hyperboloids.

Notes: (a) When equal bevel gears (having equal teeth) connect two shafts whose axes are mutually perpendicular, then the bevel gears are known as mitres.

(b) A hyperboloid is the solid formed by revolving a straight line about an axis (not in the same plane), such that every point on the line remains at a constant distance from the axis.

(c) The worm gearing is essentially a form of spiral gearing in which the shafts are usually at right angles.



2. According to the peripheral velocity of the gears: The gears, according to the peripheral velocity of the gears may be classified as:

(a) Low velocity, (b) Medium velocity, and (c) High velocity.

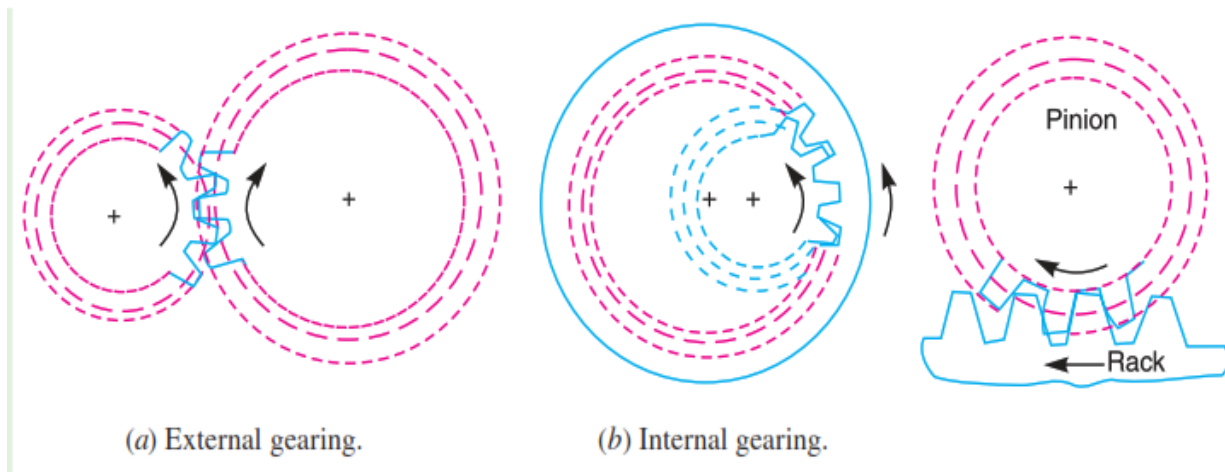
The gears having velocity less than 3 m/s are termed as low velocity gears and gears having velocity between 3 and 15 m/s are known as medium velocity gears. If the velocity of gears is more than 15 m/s, then these are called high speed gears.



3. According to the type of gearing: The gears, according to the type of gearing may be classified as:

- (a) External gearing, (b) Internal gearing, and (c) Rack and pinion.

In external gearing, the gears of the two shafts mesh externally with each other as shown in following Fig.(a). The larger of these two wheels is called spur wheel and the smaller wheel is called pinion. In an external gearing, the motion of the two wheels is always unlike, as shown in following Fig. (a).



In internal gearing, the gears of the two shafts mesh internally with each other as shown in above Fig. (b). The larger of these two wheels is called annular wheel and the smaller wheel is called pinion. In an internal gearing, the motion of the two wheels is always like, as shown in above Fig.(b).

Sometimes, the gear of a shaft meshes externally and internally with the gears in a straight line, as shown in Fig. Such type of gear is called rack and pinion. The straight-line gear is called rack and the circular wheel is called pinion. A little consideration will show that with the help of a rack and pinion, we can convert linear motion into rotary motion and vice-versa as shown in above Fig.

4. According to position of teeth on the gear surface: The teeth on the gear surface may be (a) straight, (b) inclined, and (c) curved.

We have discussed earlier that the spur gears have straight teeth where as helical gears have their teeth inclined to the wheel rim. In case of spiral gears, the teeth are curved over the rim surface.

LECTURE -39**Syllabus :- GEAR TRAINS (EXCEPT EPICYCLIC GEAR TRAIN)****GEAR TRAINS**

Sometimes, two or more gears are made to mesh with each other to transmit power from one shaft to another. Such a combination is called gear train or train of toothed wheels. The nature of the train used depends upon the velocity ratio required and the relative position of the axes of shafts. A gear train may consist of spur, bevel or spiral gears.

Types of Gear Trains

Following are the different types of gear trains, depending upon the arrangement of wheels:

1. Simple gear train, 2. Compound gear train, 3. Reverted gear train, & 4. Epicyclic gear train.

In the first three types of gear trains, the axes of the shafts over which the gears are mounted are fixed relative to each other. But in case of epicyclic gear trains, the axes of the shafts on which the gears are mounted may move relative to a fixed axis.

Simple Gear Train

When there is only one gear on each shaft, as shown in following Fig. 13.1, it is known as simple gear train. The gears are represented by their pitch circles.

When the distance between the two shafts is small, the two gears 1 and 2 are made to mesh with each other to transmit motion from one shaft to the other, as shown in following Fig. (a). Since the gear 1 drives the gear 2, therefore gear 1 is called the driver and the gear 2 is called the driven or follower. It may be noted that the motion of the driven gear is opposite to the motion of driving gear.

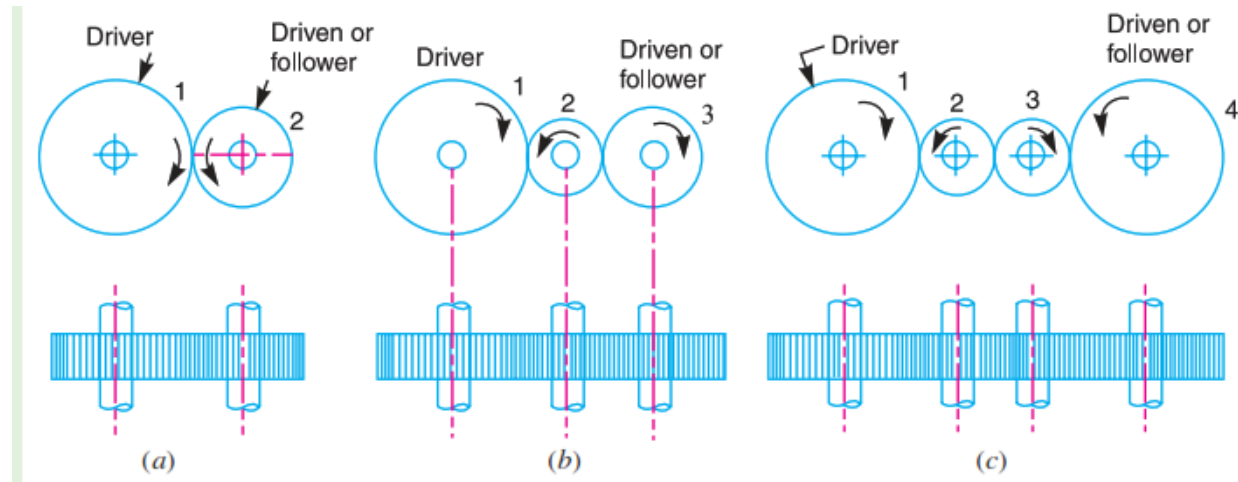
Let

N_1 = Speed of gear 1 (or driver) in r.p.m.,

N_2 = Speed of gear 2 (or driven or follower) in r.p.m.,

T_1 = Number of teeth on gear 1, and

T_2 = Number of teeth on gear 2.



Since the speed ratio (or velocity ratio) of gear train is the ratio of the speed of the driver to the speed of the driven or follower and ratio of speeds of any pair of gears in mesh is the inverse of their number of teeth, therefore

$$\text{Speed Ratio} = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

It may be noted that ratio of the speed of the driven or follower to the speed of the driver is known as train value of the gear train. Mathematically,

$$\text{Train Value} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$

From above, we see that the train value is the reciprocal of speed ratio.

Sometimes, the distance between the two gears is large. The motion from one gear to another, in such a case, may be transmitted by either of the following two methods:

1. By providing the large sized gear, or 2. By providing one or more intermediate gears.

A little consideration will show that the former method (i.e. providing large sized gears) is very inconvenient and uneconomical method; whereas the latter method (i.e. providing one or more intermediate gear) is very convenient and economical.

It may be noted that when the number of intermediate gears are odd, the motion of both the gears (i.e. driver and driven or follower) is like as shown in above Fig. (b).

But if the number of intermediate gears are even, the motion of the driven or follower will be in the opposite direction of the driver as shown in above Fig. (c).

Now consider a simple train of gears with one intermediate gear as shown in above Fig. (b).

Let

N_1 = Speed of driver in r.p.m.,

N_2 = Speed of intermediate gear in r.p.m.,

N_3 = Speed of driven or follower in r.p.m.,

T_1 = Number of teeth on driver,

T_2 = Number of teeth on intermediate gear, and

T_3 = Number of teeth on driven or follower

Since the driving gear 1 is in mesh with the intermediate gear 2, therefore speed ratio for these two gears is

$$\frac{N_1}{N_2} = \frac{T_2}{T_1} \quad \dots (i)$$

Similarly, as the intermediate gear 2 is in mesh with the driven gear 3, therefore speed ratio for these two gears is

$$\frac{N_2}{N_3} = \frac{T_3}{T_2} \quad \dots (ii)$$

The speed ratio of the gear train as shown in Fig. 13.1 (b) is obtained by multiplying the equations (i) and (ii).

$$\frac{N_1}{N_2} \times \frac{N_2}{N_3} = \frac{T_2}{T_1} \times \frac{T_3}{T_2} \text{ or } \frac{N_1}{N_3} = \frac{T_3}{T_1}$$

$$\begin{aligned} \text{Speed ratio} &= \frac{\text{Speed of driver}}{\text{Speed of driven}} = \frac{\text{No. of teeth on driven}}{\text{No. of teeth on driver}} \\ \text{Train value} &= \frac{\text{Speed of driven}}{\text{Speed of driver}} = \frac{\text{No. of teeth on driver}}{\text{No. of teeth on driven}} \end{aligned}$$

Similarly, it can be proved that the above equation holds good even if there are any number of intermediate gears. From above, we see that the speed ratio and the train value, in a simple train of gears, is independent of the size and number of intermediate gears. These intermediate gears are called idle gears, as they do not affect the speed ratio or train value of the system.

LECTURE -40**Syllabus :- GEAR TRAINS (EXCEPT EPICYCLIC GEAR TRAIN)****Compound Gear Train**

When there are more than one gear on a shaft, as shown in following Fig., it is called a compound train of gear.

We know that the idler gears, in a simple train of gears do not affect the speed ratio of the system. But these gears are useful in bridging over the space between the driver and the driven.

But whenever the distance between the driver and the driven or follower has to be bridged over by intermediate gears and at the same time a great (or much less) speed ratio is required, then the advantage of intermediate gears is intensified by providing compound gears on intermediate shafts. In this case, each intermediate shaft has two gears rigidly fixed to it so that they may have the same speed. One of these two gears meshes with the driver and the other with the driven or follower attached to the next shaft as shown in following Fig.

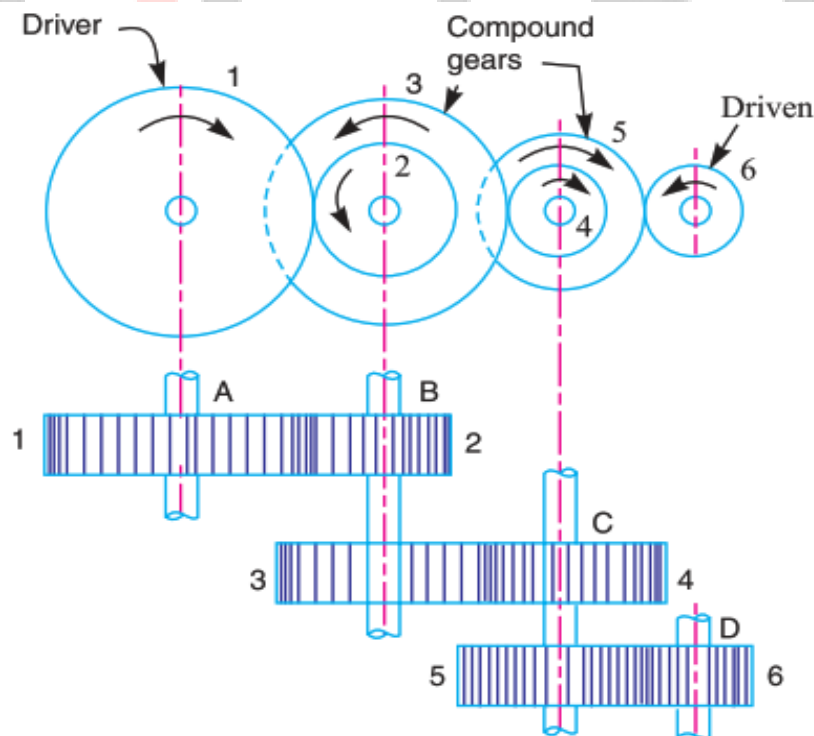


Fig.: Compound Gear Train

In a compound train of gears, as shown in above Fig., the gear 1 is the driving gear mounted on shaft A , gears 2 and 3 are compound gears which are mounted on shaft B. The gears 4 and 5 are also compound gears which are mounted on shaft C and the gear 6 is the driven gear mounted on shaft D.

Let

N_1 = Speed of driving gear 1,

T_1 = Number of teeth on driving gear 1,

$N_2, N_3, \dots, N_6$ = Speed of respective gears in r.p.m., and

$T_2, T_3, \dots, T_6$ = Number of teeth on respective gears.

Since gear 1 is in mesh with gear 2, therefore its speed ratio is

$$\frac{N_1}{N_2} = \frac{T_2}{T_1} \quad \dots (i)$$

Similarly, for gears 3 and 4, speed ratio is

$$\frac{N_3}{N_4} = \frac{T_4}{T_3} \quad \dots (ii)$$

and for gears 5 and 6, speed ratio is

$$\frac{N_5}{N_6} = \frac{T_6}{T_5} \quad \dots (iii)$$

The speed ratio of compound gear train is obtained by multiplying the equations (i), (ii), (iii),

$$\frac{N_1}{N_2} \times \frac{N_3}{N_4} \times \frac{N_5}{N_6} = \frac{T_2}{T_1} \times \frac{T_4}{T_3} \times \frac{T_6}{T_5} \text{ or } \frac{N_1}{N_6} = \frac{T_2 \times T_4 \times T_6}{T_1 \times T_3 \times T_5}$$

$$\begin{aligned} \text{i.e., Speed ratio} &= \frac{\text{Speed of the first driver}}{\text{Speed of the last driven or follower}} \\ &= \frac{\text{Product of the number of teeth on the drivers}}{\text{Product of the number of teeth on the driven}} \end{aligned}$$

$$\begin{aligned} \text{and Train value} &= \frac{\text{Speed of the first driver}}{\text{Speed of the last driven or follower}} \\ &= \frac{\text{Product of the number of teeth on the drivers}}{\text{Product of the number of teeth on the driven}} \end{aligned}$$

Reverted Gear Train

When the axes of the first gear (i.e. first driver) and the last gear (i.e. last driven or follower) are co-axial, then the gear train is known as reverted gear train as shown in following Fig.

We see that gear 1 (i.e. first driver) drives the gear 2 (i.e. first driven or follower) in the opposite direction. Since the gears 2 and 3 are mounted on the same shaft, therefore they form a compound gear and the gear 3 will rotate in the same direction as that of gear 2. The gear 3 (which is now the second driver) drives the gear 4 (i.e. the last driven or follower) in the same direction as that of gear 1. Thus, we see that in a reverted gear train, the motion of the first gear and the last gear is like.

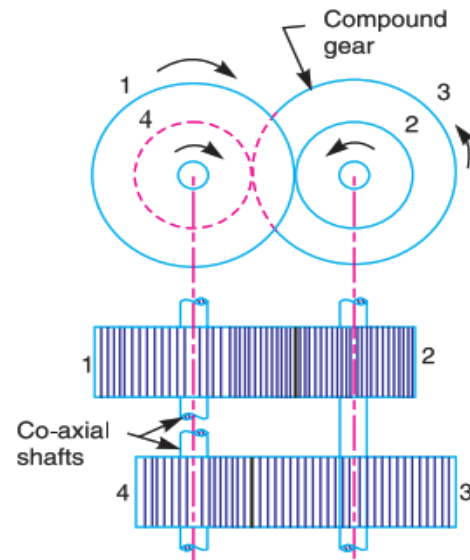


Fig.: Reverted Gear Train

Let

T_1 = Number of teeth on gear 1,

r_1 = Pitch circle radius of gear 1, and

N_1 = Speed of gear 1 in r.p.m.

Similarly,

T_2, T_3, T_4 = Number of teeth on respective gears,

r_2, r_3, r_4 = Pitch circle radii of respective gears, and

N_2, N_3, N_4 = Speed of respective gears in r.p.m.

Since the distance between the centres of the shafts of gears 1 and 2 as well as gears 3 and 4 is same, therefore

$$r_1 + r_2 = r_3 + r_4 \quad \dots (i)$$

Also, the circular pitch or module of all the gears is assumed to be same, therefore number of teeth on each gear is directly proportional to its circumference or radius.

$$\therefore T_1 + T_2 = T_3 + T_4 \quad \dots (ii)$$

and $\text{Speed ratio} = \frac{\text{Product of number of teeth on driven}}{\text{Product of number of teeth on drivers}}$

or
$$\frac{N_1}{N_4} = \frac{T_2 \times T_4}{T_1 \times T_3} \quad \dots (iii)$$

From equations (i), (ii) and (iii), we can determine the number of teeth on each gear for the given centre distance, speed ratio and module only when the number of teeth on one gear is chosen arbitrarily.

The reverted gear trains are used in automotive trans-missions, lathe back gears, industrial speed reducers, and in clocks (where the minute and hour hand shafts are co-axial).

